WORKSHEET

Comparing data

1 The ages (in years) of male and female employees in a small business are shown below.

Male:	27	42	29	35	21	19	33	26	34	51	38
	47	25	60								
Female:	18	32	26	25	21	24	33	26			

a Display the data in a back-to-back stem-and-leaf plot.

b Comment on the shape of the data for males and females.

c State the median and interquartile range for males.

d State the median and interquartile range for females.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 1 16/09/15 10:09 AM

e Using these summary statistics, compare the ages of male and female employees in this business.

2 The amounts of money raised (in dollars) by students in two Year 12 classes in a school charity event are as follows.

12X:	15	11	25	41	30	1	20	22	5	7	4	18
	13	11	10	33	13	38						
12Y:	3	5	16	22	14	12	8	9	2	21	30	18
	11	15	13	18	21	16	10					

a Display the data in a back-to-back stem-and-leaf plot.

b Comment on the shape of the data for each class.

c Calculate the median, range and interquartile range for 12X.

d Calculate the median, range and interquartile range for 12Y.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 2 16/09/15 10:09 AM

e Using these summary statistics, compare the amounts of money raised by individual students in the two classes.

3 The amount of weight lost over a period of 6 months by participants in two different weight-loss programs is recorded below.

Program A:	5	9	4	3	15	7	21	6	11	10
	8	8	6	7	11	9	17			
Program B:	3	6	7	2	5	4	6	3	1	5
	2	6	4	5	2	6	5	7	5	1
	2									

a Calculate the five-number summary for Program A.

b Calculate the five-number summary for Program B.

c Put the two sets of data in parallel boxplots.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 3 16/09/15 10:09 AM

d Comment on the shape of the data for each group.

e Use the summary statistics and boxplots to compare the two groups and state which weight-loss program appears to be more effective.

4 A group of men and women were surveyed on the amount of money (to the nearest dollar) they had spent that week on groceries. The data obtained are as follows.

Women:	0	0	0	18	35	42	58	60	84	115	
	122	142	142	176	179	188	197	214	280		
Men:	0	0	0	0	0	0	0	0	12	23	39
	57	97	148	176	255						

a Calculate the five-number summary for the women.

b Calculate the five-number summary for the men.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 4 16/09/15 10:09 AM

c Put the two sets of data in parallel boxplots.

d Comment on the shape of the data for each group.

e Use the summary statistics and boxplots to compare the two groups and state which group spends more on groceries.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 5 16/09/15 10:09 AM

5 The heights (in cm) of the players in the Open Boys basketball team and the Open Girls basketball team at Doontown High School were recorded.

Boys:	185	179	189	177	191	192	188	174	193
Girls:	167	177	172	181	176	164	183	179	180

a Display the data in a back-to-back stem-and-leaf plot.

b Comment on the shape of the data for each team.

c Calculate the mean and standard deviation for the boys' team.

d Calculate the mean and standard deviation for the girls' team.

e Using these summary statistics, compare the heights of the players in the two teams.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 6 16/09/15 10:09 AM

6 The weights (to the nearest kg) of a group of children on their second, fourth and sixth birthdays were recorded as follows.

2 year olds:	12	11	14	16	13	12	11	10	9	10	13
4 year olds:	16	13	17	14	19	21	16	15	14	18	15
6 year olds:	17	25	22	30	31	24	22	28	28	27	25

a Calculate the five-number summary for the three different ages.

b Display the data in parallel boxplots.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 7 16/09/15 10:09 AM

c Comment on the shape of the data for each group.

d Calculate the range and interquartile range for each age group.

e Using the summary statistics, compare the three groups of weights. Can you conclude that there is an association between the two variables?

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 8 16/09/15 10:09 AM

Answers

1 a

Ages of males	Stem	Ages of females
9	1	8
9 7 6 5 1	2	1 4 5 6 6
8 5 4 3	3	2 3
7 2	4	
1	5	
0	6	

Key: 1|8 means 18

b Males: positively skewed; females: symmetrical.

c Median = 33.5, IQR = 16

d Median = 25.5, IQR = 6.5

e Female employees tend to be younger, as the median of this group is lower than that for males. The ages of the female employees are also less spread out than the ages of the males. This is supported by the lower interquartile range for the females.

2 a

12X	Stem	12Y
7 5 4 1	0	2 3 5 8 9
8 5 3 3 1 1 0	1	0 1 2 3 4 5 6 6 8 8
5 2 0	2	1 1 2
8 3 0	3	0
1	4	

Key: 1|8 means 18

b Both groups of data are positively skewed.

c Median = 14, range = 40, IQR = 15

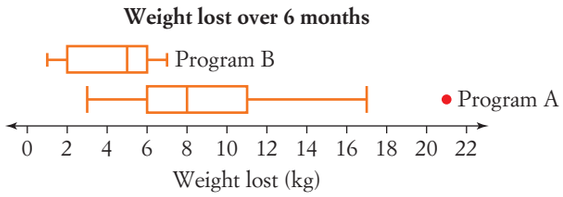
d Median = 14, range = 28, IQR = 9

e The median amount of money raised by individual students in both classes is the same. The range and interquartile range are both larger for 12X than for 12Y, indicating the amounts of money raised had a much greater spread in 12X.

3 a Minimum = 3, Q_1 = 6, median = 8, Q_3 = 11, maximum = 17, outlier = 21

b Minimum = 1, Q_1 = 2, median = 5, Q_3 = 6, maximum = 7

c



© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 9 16/09/15 10:09 AM

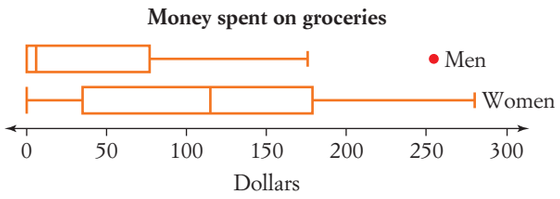
d The data for Program A is positively skewed with an outlier, whereas the data for Program B is slightly negatively skewed.

e The interquartile ranges for the two groups are quite similar (5 for Program A and 4 for Program B), the ranges are very different (18 for Program A and 6 for Program B). The results for Program A have a much greater spread than for Program B, indicating that Program B gives more consistent results. However, Q_1 for Program A = Q_3 for Program B, which indicates that 75% of participants in Program A lost more weight than 75% of participants in Program B. Program A appears to be more effective overall.

4 a Minimum = 0, Q_1 = 35, median = 115, Q_3 = 179, maximum = 280

b Minimum = 0, Q_1 = 0, median = 6, Q_3 = 97, maximum = 176, outlier = 255

c



d The data for men is positively skewed with an outlier, whereas the data for women is approximately symmetrical.

e Looking at the boxplots, we can see that women spend more money on groceries than do men. Whereas the range of the data in both groups is similar, there is more than $100 dollars difference between the two medians, and the first quartile for women is $29 more than the median for men.

5 a

Boys	Stem	Girls
	16	4 7
9 7 4	17	2 6 7 9
9 8 5	18	0 1 3
3 2 1	19	

Key: 4|17 means 174 cm

b The data for boys is evenly distributed. The data for girls is symmetrical.

c $\bar{x}$ = 185.33… ≈ 185.3 cm, s_x = 7.01… ≈ 7.0 cm

d $\bar{x}$ = 175.44… ≈175.4 cm, s_x = 6.50… ≈ 6.5 cm

e The mean height of the boys' team is almost 10 cm greater than that of the girls' team; however, the standard deviations of the two groups are quite close. Overall, this tells us that the boys are generally taller, but the spread of the heights of the two teams is similar.

6 a 2 year olds: minimum = 9, Q_1 = 10, median = 12, Q_3 = 13, maximum = 16

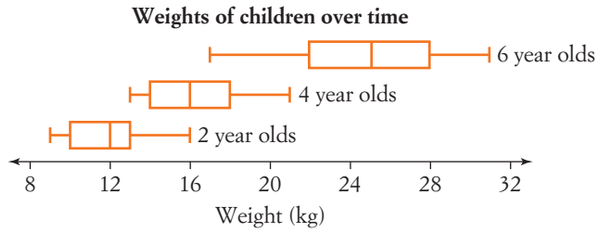
© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00057.indd 10 16/09/15 10:09 AM

4 year olds: minimum = 13, Q_1 = 14, median = 16, Q_3 = 18, maximum = 21

6 year olds: minimum = 17, Q_1 = 22, median = 25, Q_3 = 28, maximum = 31

b



c The data for each group is approximately symmetrical.

d 2 year olds: range = 7, IQR = 3

4 year olds: range = 8, IQR = 4

6 year olds: range = 14, IQR = 6

e As the children get older, their weights increase, and also the range and IQR of their weights increase. From this, we can conclude that there is an association between the two variables, age and weight.

© Cengage Learning Australia Pty Ltd 2015